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Asia Communication Infrastructure

800VDC Architecture for Next-Generation AI Data Centers



CITI'S TAKE

In this Super-Sector report, we outline likely implications of the 800VDC transition and potential Asia beneficiaries in the grey space and IT rack power/cooling in AI datacenters. As GPU rack power density approaches 1MW, we forecast 800VDC to reach 79% of new global DC capacity by 2030E, with SST demand surging from 901 MVA in 2027E to 37,152 MVA in 2030E to replace traditional LV transformers. 800VDC architecture is driving combined BESS and BBU battery demand at an 80%+ CAGR through 2030E. We see this as a multi-layer investment opportunity across six segments: DC operators riding AI demand tailwind; power equipment names riding SST architecture upcycle; BESS and BBU suppliers benefiting from rising battery demand; IT rack power/cooling vendors (incl. BBU, MLCC) benefiting from surging power density and cooling requirement; liquid cooling providers capturing structural thermal upgrades; and power semiconductor suppliers underpinning the entire conversion chain.

Traditional AC Chain Gives Way to SST — GPU rack power density has grown nearly 100x since 2020, rendering conventional AC distribution physically unviable. We project 800VDC adoption to reach 79% of new global data center capacity by 2030E, displacing legacy power architecture while creating new equipment categories including SST, BESS, BBU, and new cooling solutions.

Data Center Operators — We expect the near-term AI demand tailwind to be intact. We like China DC market leaders **GDS** and **VENT**. In Australia, we think **NextDC** and **CDC/Infratil** are positioned to incorporate 800VDC into future builds.

Power Equipment — We see **Delta** as uniquely positioned with end-to-end grid-to-chip portfolio. **Hyosung**, **Sieyuan** and **TGOOD** should benefit from LV transformer and SST for domestic buildout and export opportunities. **Hongfa** should ride rising relay products for 800VDC migration. **Hitachi Energy India**, **GE Vernova T&D India**, **CG Power and Industrial** should benefit from India domestic demand and exports.

BESS — 800VDC battery demand could accelerate at an 80%+ CAGR in 2027-30E; **Sungrow** (SST launch July 2026), **BYD**, **CATL** and **EVE Energy** are key beneficiaries.

IT Rack Power & Cooling — We forecast BBU content value per rack to rise from US\$4-5K in 2022 to US\$33-34K by 2029E, with **Panasonic**, **Dynapack** and **AES** the direct plays. We see **Murata**, **SEMCO** and **Yageo** as key MLCC beneficiaries. We also believe Taiwan and Chinese rack cooling players may benefit from IT cabinets and liquid cooling upgrade.

Cooling Solutions — Higher rack power densities make liquid cooling a structural necessity. **LG Electronics**, **Daikin** and **Envicool** are key names to watch.

Power Semiconductors — **VIS** leads at high-voltage front end; **UMC** leveraged to PMIC/BCD specialty nodes; **Renesas** could benefit via Transphorm GaN acquisition. For China, **SG Micro**, **CR Micro** and **Silan Micro** could benefit from SiC/GaN demand.

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See Appendix A-1 for Analyst Certification, Important Disclosures and Research Analyst Affiliations.

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800VDC Architecture for Next-Generation AI Data Centers

In this report, we outline likely implications of the 800V DC transition in AI infrastructure and potential beneficiaries in Asia.

The rapid proliferation of AI workloads is fundamentally reshaping the power infrastructure of modern data centers. GPU rack power density has grown nearly 100-fold compared to traditional web servers — from 10kW per rack in 2020, to 120kW with Blackwell, and should exceed 1MW with the forthcoming Rubin Ultra and Kyber platforms by 2028. This near-exponential trajectory does not merely cause an incremental engineering challenge; it represents a structural inflection point that renders conventional power distribution architectures physically and economically unviable.

In response, the industry is converging on 800VDC as the optimal power distribution standard for next-generation AI infrastructure. By delivering power at higher voltage, 800VDC dramatically reduces current levels, copper requirements, and conversion losses across the entire power chain from the utility grid to the compute chip.

Figure 1. 800VDC Architecture Key Potential Asia Beneficiaries (data as at 16-Jul-2026)

Company	Ticker	Rating	Analyst	TP	Price	Currency	Mar Cap	P/S (x)	P/E (x)	EPS Growth	Sales Growth	PIBV (x)	ROE	Div. Yield
					16-Jul		(US\$M)	2026	2027	2025	2026	2027	2026	2027
Data Centers														
GDS	GDS.O	Buy	Kyna Wong	58.10	32.75	USD	6,759	4.0	4.0	217%	101%	-106%	11%	12%
VNET	VNET.O	Buy	Kyna Wong	16.40	7.72	USD	2,198	1.2	1.0	-207%	-175%	92%	20%	18%
NextDC	NXT.AX	Buy	Siraj Ahmed	19.10	13.45	AUD	7,164	18.5	14.7	-10%	-98%	-207%	6%	14%
Goodman	GMG.AX	Buy	Howard Penny	40.00	29.12	AUD	41,758			11%	11%		2.4	2.2
Infratil	IFT.AX	Buy	Suraj Nebhani, CFA	15.18	12.85	AUD	9,031	4.5	4.6	-132%	280%	-143%	23%	-9%
Stockland	SGP.AX	Buy	Suraj Nebhani, CFA	5.10	4.08	AUD	6,958			11%	11%		0.9	0.9
Power Equipment														
Delta	2308.TW	Buy	Laura (Chia Yi) Chen	2,480.00	1,905.00	TWD	154,921	6.0	4.2	48.6	30.9	71%	69%	57%
Hongfa	600885.SS	Buy	Jamie Wang	43.00	33.00	CNY	7,544	2.4	2.0	24.8	19.8	-22%	11%	25%
Inovance	300124.SZ	Buy	Jamie Wang	75.00	60.31	CNY	24,123	3.0	2.5	28.1	22.8	17%	15%	23%
Sieyuan Electric	002028.SZ	Buy	Pierre Lau, CFA	216.00	151.40	CNY	17,503	4.2	3.3	31.8	24.4	53%	18%	31%
TBEA	600089.SS	Buy	Pierre Lau, CFA	36.00	18.88	CNY	14,093	0.9	0.8	10.5	9.2	84%	23%	14%
TGOOD	300001.SZ	Buy	Air Ma	49.60	31.28	CNY	4,877	2.0	1.7	22.8	17.4	36%	17%	31%
Dongfang Electric	1072.HK	Buy	Pierre Lau, CFA	30.00	21.22	HKD	9,357	0.8	0.7	13.1	10.6	18%	26%	24%
Yantai Jereh	002353.SZ	Buy	Desmond Law	146.00	130.50	CNY	19,738	6.4	5.0	37.0	27.0	2%	35%	37%
Weichai Power	2338.HK	Buy	Jeff Chung	50.00	32.04	HKD	35,296	0.9	0.8	14.6	11.5	-4%	50%	27%
Mitsubishi Electric	6503.T	Buy	Masahiro Shibano	7,600.00	5,666.00	JPY	71,852	2.0	1.8	28.6	21.9	15%	27%	30%
Hyundai Electric	267260.KS	Buy	Pierre Lau, CFA	1,440,000.00	797,000.00	KRW	18,975	5.8	4.7	28.3	22.0	46%	38%	29%
Hyosung Heavy	298040.KS	Buy	Pierre Lau, CFA	4,500,000.00	2,789,000.00	KRW	17,176	3.6	2.9	30.7	22.2	167%	42%	38%
LS Electric	010120.KS	Buy	Pierre Lau, CFA	320,000.00	190,000.00	KRW	18,823	4.5	3.5	60.3	41.7	16%	68%	45%
Hitachi Energy India	HITN.NS	Buy	Anusha Madireddy	46,700.00	33,545.00	INR	15,851	18.4	11.4	151.4	83.2	123%	157%	82%
GE Vernova T&D India	GETD.NS	Buy	Anusha Madireddy	6,200.00	4,584.10	INR	12,444	18.8	14.0	94.8	62.1	236%	103%	53%
CG Power	CGPO.NS	Buy	Anusha Madireddy	1,100.00	921.95	INR	15,394	11.7	9.0	121.7	78.2	-30%	19%	56%
Siemens Energy India	SIEE.NS	Neutral	Anusha Madireddy	4,000.00	3,405.30	INR	12,857	10.9	9.2	70.6	56.2	83%	56%	26%
BESS														
BYD	1211.HK	Buy	Jeff Chung	142.00	90.95	HKD	105,725	0.7	0.6	17.8	15.1	-22%	24%	18%
CATL (H)	3750.HK	Buy	Jack Shang, CFA	888.00	621.00	HKD	255,764	3.7	3.0	23.8	19.3	40%	40%	23%
CATL (A)	300750.SZ	Buy	Jack Shang, CFA	603.00	366.20	CNY	255,764	2.5	2.0	16.2	13.2	40%	40%	23%
Sungrow	300274.SZ	Buy	Pierre Lau, CFA	185.00	105.84	CNY	32,415	2.0	1.6	14.6	11.6	23%	11%	26%
EVE Energy	300014.SZ	Buy	Jack Shang, CFA	87.90	51.61	CNY	16,569	1.0	0.7	12.8	10.4	1%	100%	23%
IT Rack Power & Cooling														
Lenovo	0992.HK	Buy	Kyna Wong	31.00	22.26	HKD	35,206	0.4	0.4	16.9	14.0	34%	44%	21%
ZTE	0763.HK	Neutral	Kyna Wong	25.40	26.28	HKD	16,028	0.8	0.7	15.1	12.9	-33%	28%	17%
BYD Electronics	0285.HK	Sell	Kyna Wong	22.60	23.46	HKD	6,740	0.3	0.2	13.6	9.2	-18%	-5%	47%
Dynapack	3211.TWO	Buy	Angela Hsu	800.00	426.50	TWD	2,060	3.7	2.7	28.4	17.1	-49%	66%	67%
AES	6781.TW	Buy	Angela Hsu	1,680.00	1,045.00	TWD	2,795	4.7	3.7	22.5	17.0	50%	21%	32%
Yageo	2327.TW	Buy	Michael Hung	1,500.00	776.00	TWD	50,326	9.3	6.6	43.6	24.7	20%	55%	76%
SEMCO	009150.KS	Buy	Peter Lee	3,000,000.00	1,277,000.00	KRW	62,997	6.2	3.9	60.2	28.3	4%	124%	113%
Murata	6981.T	Buy	Takayuki Naito	15,000.00	8,400.00	JPY	94,751	8.4	7.8	66.1	44.1	30%	2%	50%
Panasonic	6752.T	Buy	Masahiro Shibano	5,100.00	4,218.00	JPY	61,030	1.2	1.2	60.6	21.5	-18%	-56%	181%
Cooling Solution														
LGE	066570.KS	Buy	Peter Lee	400,000.00	179,000.00	KRW	19,257	0.3	0.2	9.2	7.1	230%	253%	29%
Dakin	6367.T	Buy	Graeme McDonald	29,500.00	24,870.00	JPY	42,894	1.5	1.3	26.5	23.3	2%	4%	14%
Envicool	002837.SZ	Sell	Eric Lau	46.15	67.86	CNY	12,775	10.9	8.7	109.1	87.9	-12%	58%	24%
Power Semiconductors														
VIS	5347.TWO	Buy	Laura (Chia Yi) Chen	170.00	187.50	TWD	11,033	5.9	4.6	31.6	25.3	0%	39%	25%
UMC	2303.TW	Buy	Laura (Chia Yi) Chen	88.00	160.00	TWD	63,001	7.4	6.6	33.0	27.8	-12%	45%	19%
Renesas	6723.T	Neutral	Takero Fujiwara	5,200.00	4,139.00	JPY	46,534	4.7	4.4	19.1	17.8	-11%	21%	7%
SG Micro	300661.SZ	Buy	Kevin Chen	120.00	107.32	CNY	9,876	13.5	11.2	75.3	54.3	9%	61%	39%
CR Micro	686396.SS	Neutral	Kevin Chen	47.00	65.50	CNY	12,852	7.1	6.5	54.8	49.1	50%	38%	12%
Silan Micro	600460.SS	Neutral	Kevin Chen	30.00	36.22	CNY	8,904	3.9	3.4	47.6	37.8	216%	82%	26%

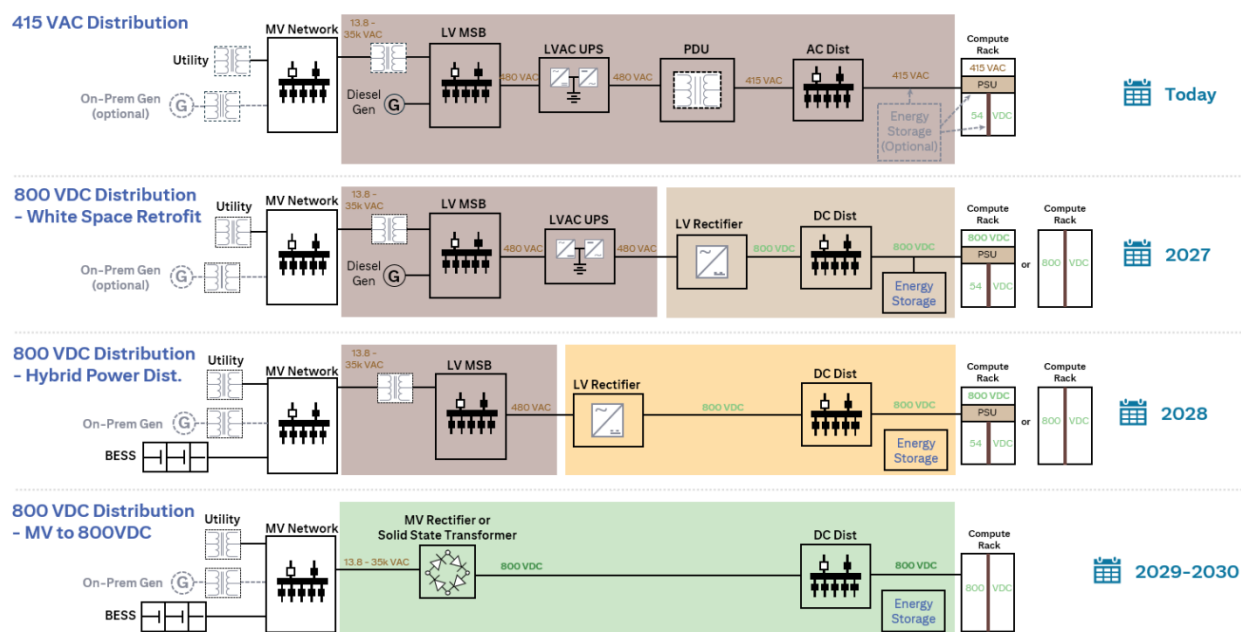
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Source: dataCentral, Citi Research

Nvidia's 800VDC Architecture: Four Generations, from Multi-Stage AC to a Single Integrated DC Conversion

NVIDIA's reference architecture defines 4 generations of data center power distribution, each representing a step-change reduction in conversion stages (Figure 1). Today's incumbent 415VAC architecture routes power through five or more sequential stages (transformer, LV switchboard, AC UPS, PDU, and in-rack PSU) before reaching the compute chip at 54VDC. The first transition step, 800VDC White Space Retrofit, inserts an LV Rectifier and a dedicated HVDC Power Sidecar rack into existing facilities, converting AC to 800VDC at the row level without disturbing grey-space infrastructure. The subsequent Hybrid Power Distribution phase moves rectification further upstream, eliminates the central AC UPS, and distributes 800VDC across the full data hall via a common DC busway. The terminal state — MV to 800VDC via solid-state transformer (SST) — eliminates the LV transformer layer entirely, converting grid-level medium voltage (e.g., 35kVAC) directly to 800VDC in a single device rated up to 7.5MVA at $\geq 98.5\%$ efficiency.

Figure 2. Nvidia 800 VDC Architecture for Next-Generation AI Infrastructure



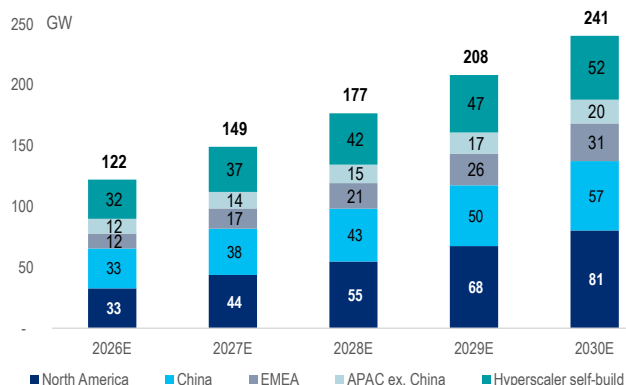
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Source: Nvidia, Citi Research

Citi Research's proprietary projection for global data center buildout shows total capacity in service growing from **122GW in 2026E to 241GW by 2030E**. North America and China are the primary buildout destinations, with hyperscaler self-build capacity surging alongside the buildout.

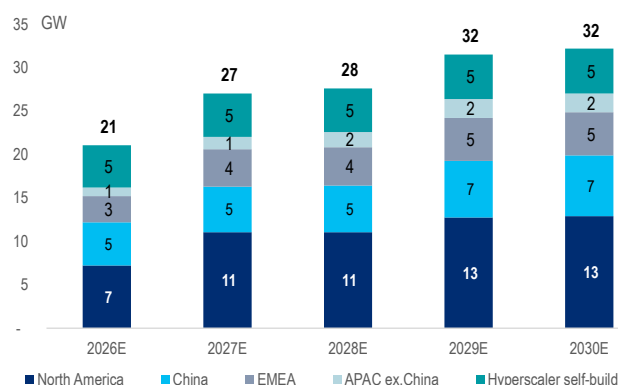
On an incremental basis, Citi projects annual new capacity additions at **21-32GW per year from 2026E through 2030E**, suggesting the build cycle is entering a sustained high-volume phase rather than a single-year spike. North America contributes 7-13GW of incremental capacity annually, China 5-7GW, and hyperscaler self-build 3-5GW.

Figure 3. Total DC Capacity in Services (GW)



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Source: Citi Research Estimates

Figure 4. Incremental DC Capacity in Services (GW)

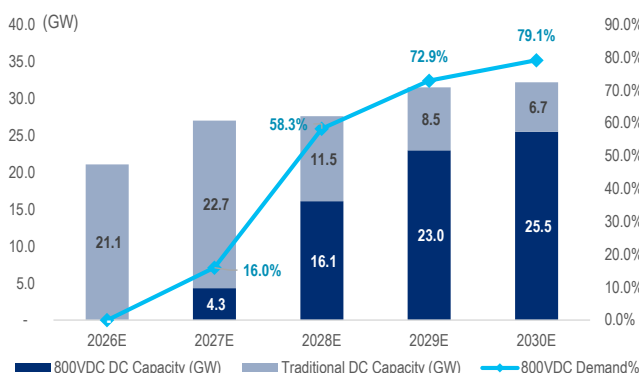


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Source: Citi Research Estimates

From a GPU chipset requirement perspective, demand for 800VDC-enabled capacity is rising steeply as a share of total new builds, underpinned by [Nvidia's Rubin Ultra roadmap remaining intact](#). We project 800VDC's adoption demand based on the GPU rack deployment requirement at 16.0% in 2027E, accelerating to **58.3%/72.9%/79.1% in 2028E/29E/30E**, reflecting the rising demand of the HVDC power rack driven by the commercial launch of 800VDC GPU platforms.

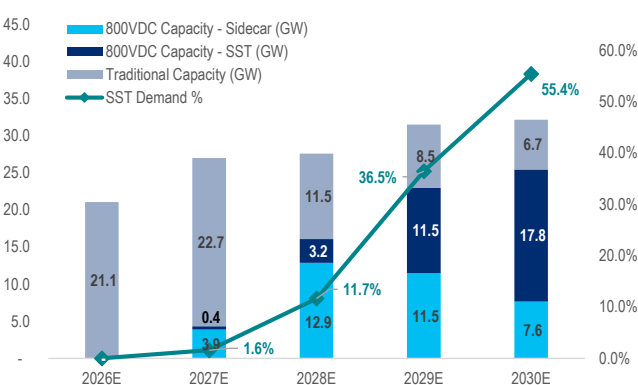
In the 800VDC adoption transition roadmap, sidecar dominates the early phase, with 3.9GW in 2027E and 12.9GW in 2028E, as existing facilities are retrofitted at the row level without large-scale grey-space reconstruction. We estimate SST-based builds could start in 2028E and accelerate to 11.5GW in 2029E and 17.8GW in 2030E, with SST adoption demand of total new capacity reaching 55.4% by 2030E.

Figure 5. Global Data Center Build-out with 800VDC Adoption Requirement by GPU Chip



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Source: Citi Research Estimates

Figure 6. Global Data Center Build-out by Architecture Type

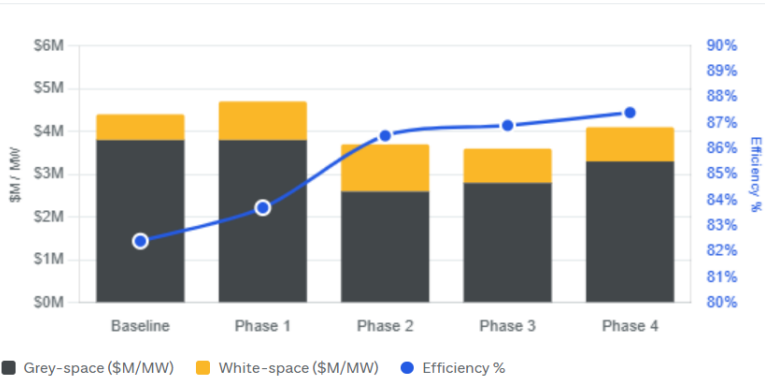


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Source: Citi Research Estimates

On the cost side, according to SemiAnalysis' projection, the sidecar retrofit phase carries a modest near-term premium of +US\$0.40M/MW versus today's baseline. Going forward, with the transition to SST deployment, the total electrical cost per MW falls by US\$0.70M versus baseline as the central UPS, LV transformer, and LV switchgear are eliminated.

Figure 7. Total Electrical Content per MW Could Slightly Drop

\$M/MW (Grey + White space) - Right axis: System Efficiency %



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Source: SemiAnalysis, Citi Research

Traditional AC Chain Has Four Lossy Conversion Stages — All of Which Collapse into One under 800VDC

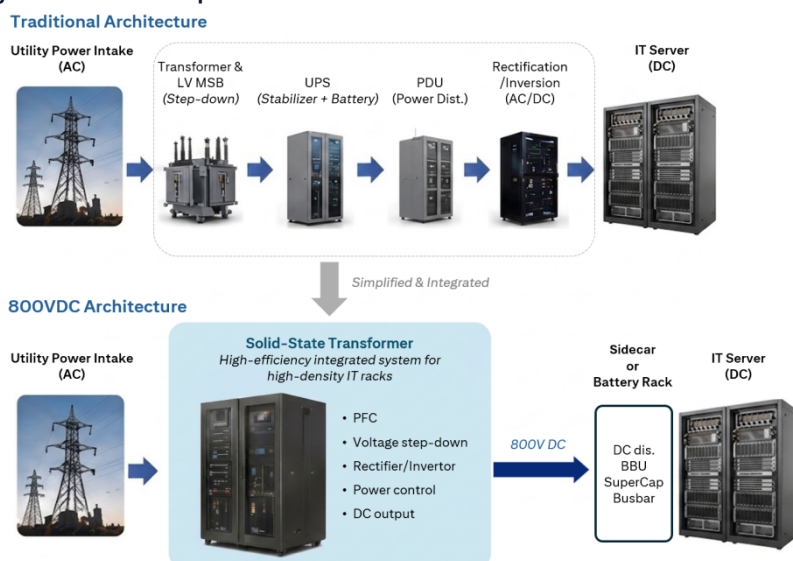
We delve into AI data centers' power structure in grey space. The traditional architecture routes utility AC power to IT server in white space through the following major parts:

- **Transformer & MSB:** after utility power intake, transformers convert medium-voltage grid power to low-voltage AC, then LV MSB performs power distribution.
- **UPS:** provides power conditioning and short-term power backup. In double-conversion topology, the UPS performs AC→DC→AC conversion, introducing ~3–5% losses and requiring significant floor space and battery infrastructure.
- **PDU:** distributes conditioned AC power to rack rows, with metering and circuit protection.
- **Rectification/inversion in-rack:** PSU shelf inside each compute rack performs the final AC-to-DC conversion, outputting low-voltage DC to compute racks.

Besides, another important part in the data center power system is the back-up power, which is dominated by diesel generator set (GenSet) currently.

800VDC SST architecture replaces the lengthy AC chain with a single integrated device. The SST performs all necessary functions — power factor correction (PFC), voltage step-down, rectification/inversion, power control, etc — within a single compact unit. It accepts medium-voltage AC directly from the utility grid and outputs 800VDC to the DC distribution network, which feeds compute racks without further conversion until the final voltage step-down at the GPU board level. Between the SST and the IT server sits the sidecar or battery rack — a purpose-built intermediate module that serves as the intelligence and resilience layer of the 800VDC power chain.

Figure 8. Traditional power distribution architecture vs SST architecture



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Source: Citi Research

Implications of the 800VDC architecture span following structural dimensions:

- **Copper consumption saving:** Doubling the distribution voltage from 415VAC to 800VDC halves the current required to deliver the same power, which translates directly into substantially thinner, lighter cables that dramatically reduce the cost of high-grade copper throughout the facility.
- **Equipment layers eliminated:** UPS, LV transformer, PDU, and in-rack PSU shelf are all removed from the primary power path, reducing both capital cost and maintenance complexity.
- **PUE improvement:** Elimination of AC/DC rectification across a MW-scale data center improves end-to-end power delivery efficiency and lowers cooling requirement, which significantly reduce operational PUE.
- **Native energy storage integration:** Rather than a centralised UPS battery bank, energy storage is distributed natively on the 800VDC bus, creating a layered, resilient power architecture purpose-built for the dynamic demands of AI workloads. An 800VDC DC architecture also facilitates green energy sources and storage systems to connect directly to the facility power bus, which is a strategic enabler of hyperscalers with binding carbon neutrality commitments.
- **Footprint reduction:** Grey-space power equipment, which in traditional facilities can consume 30–40% of total building area, is dramatically compressed — a critical advantage as power infrastructure increasingly rivals compute in space consumption.

Key Equipment Demand Projection in 800VDC Transition

We estimate an outline of per-GW data center IT load normalised comparison of equipment capacity requirements across the two architectures, translating the qualitative architectural differences into concrete capacity demand metrics.

There are four major categories that define the traditional AC power chain — LV transformer/MSB, UPS and associate UPS battery, and PDU/RPP — will be integrated into a SST system in the 800VDC architecture. For every 1GW of new 800VDC data center IT load built, approximately 2,083 MVA of LV transformer procurement, 1,563 MW of UPS procurement, 1,500 MW of PDU procurement, and 326 MWh of UPS battery procurement are permanently forgone relative to the traditional architecture, on our estimates.

The 800VDC SST architecture introduces the following new major demand categories that have no precedents in traditional data center procurement:

- BBU power at 840 MW/GW (housed in the sidecar or in battery rack, providing high-rate millisecond-to-second buffering), BBU battery at 76 MWh/GW (the energy storage element of the rack-level buffer).
- BESS power requires 1,500 MW/GW at the facility level with BESS battery at 530 MWh/GW serving as facility-level energy storage, replacing the centralised UPS battery and adding peak shaving capability.
- A DC distribution system of 1,500MW/GW will replace the traditional PDU.
- Notably, the SST replaces the LV Transformer & MSB on a one-for-one MVA basis (2,083 MVA/GW in both architectures) — confirming that the total power conversion capacity requirement is unchanged, but the device performing that conversion is entirely different, with materially higher efficiency and functional integration.

Backup GenSet remains constant across both architectures. At 2,063 MVA/GW in both the traditional and 800VDC columns, the GenSet requirement is the sole equipment category that is genuinely architecture-neutral — driven purely by the total facility load (IT load × PUE × redundancy) rather than the AC/DC topology of the power distribution system. Backup GenSet can also be replaced by large scale energy storage batteries in the future when the economic yield becomes reasonable.

Figure 9. Key Power Equipment Capacity Requirements Per GW IT Load of Data Center Build-out

Per GW - Equipment Capacity	Traditional	800VDC SST	Unit	Note
LV Transformer/MSB	2,083	-	MVA	Substituted by SST
SST	-	2,083	MVA	Integrated SST
Backup GenSet	2,063	2,063	MVA	Can be Battery
UPS	1,563	-	MW	Eliminated
PDU	1,500	-	MW	Eliminated
DC distribution	-	1,500	MW	in Sidecar/Integrated SST
UPS Battery	326	-	MWh	Eliminated
BBU Power	-	840	MW	in Sidecar/Integrated SST
BBU Battery	-	76	MWh	in Sidecar/Integrated SST
BESS Power	-	1,500	MW	Facility level
BESS Battery	-	530	MWh	Facility level

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Source: Citi Research

SST Demand Surges as LV Transformer Market Structurally Declines

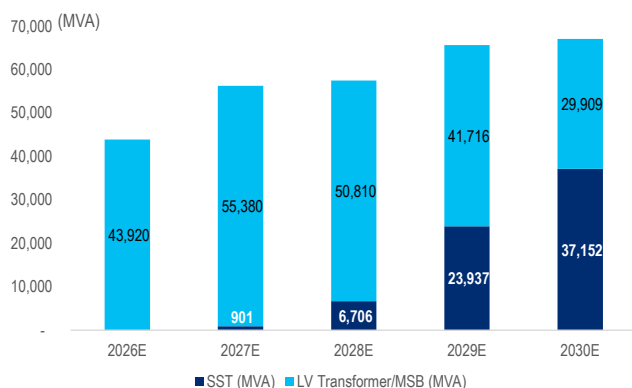
We estimate total global transformer-equivalent capacity demand (SST + LV transformer combined) could grow from ~**43,920 MVA in 2026E** to ~**67,000 MVA in 2030E**, driven by 21.1GW (2026E) to 32.2 GW (2030E) data center buildout. Within this total, the compositional shift is dramatic: LV transformer demand progressively cedes share to SST as penetration accelerates. SST demand grows from 901 MVA in 2027E to 37,152 MVA in 2030E, while LV transformer demand drops from 55,380 MVA in 2027E to 29,909 MVA in 2030E.

GenSet Capacity Demand Grows Steadily

GenSet demand is architecture-agnostic: whether a data center runs on 415VAC or 800VDC, it requires the same diesel backup generation capacity to cover full-facility load (IT + cooling + ancillary) in the event of a grid outage.

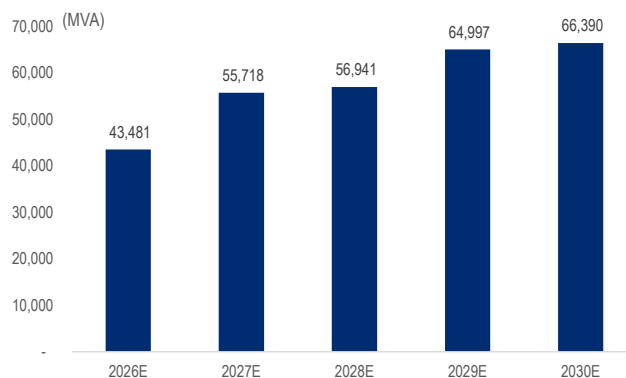
Total incremental GenSet capacity demand grows from **43,481 MVA in 2026E** to **66,390 MVA in 2030E** — a compound annual growth rate of 11% over the four-year period. This growth is a direct function of the absolute volume of new data center capacity being added globally. The implication is that GenSet manufacturers should face a straightforward, durable demand tailwind through 2030E.

Figure 10. Transformer Capacity Demand: SST to Replace LV Transformer/MSB



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Source: Citi Research Estimates

Figure 11. Backup GenSet Capacity Incremental Demand Driven by DC Buildout



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Source: Citi Research Estimates

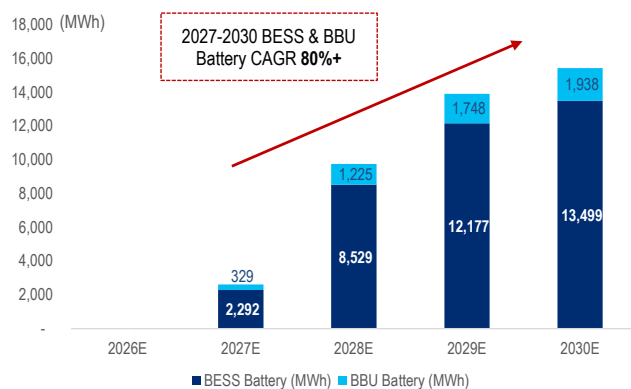
BESS and BBU Emerge as High-Growth New Categories While UPS Faces Structural Decline

We believe combined BESS and BBU battery capacity demand is negligible in 2026 (reflecting the early-stage 800VDC transition phase, which retains legacy UPS infrastructure), but should accelerate sharply from 2027 as native 800VDC deployments ramp. We estimate that BESS battery demand grows from **2,292 MWh in 2027E** to **13,499 MWh in 2030E** and BBU battery demand grows from **329 MWh in 2027E** to **1,938 MWh in 2030E**, representing a **2027–30E CAGR of 80%+** for BESS and BBU battery demand in the global data center buildout.

The functional distinction between BESS and BBU is critical to understanding their respective demand drivers. BBU — housed within the sidecar rack or battery rack — operates at the rack level in white space, providing millisecond-to-second buffering of GPU load swings via high-rate lithium batteries or supercapacitors.

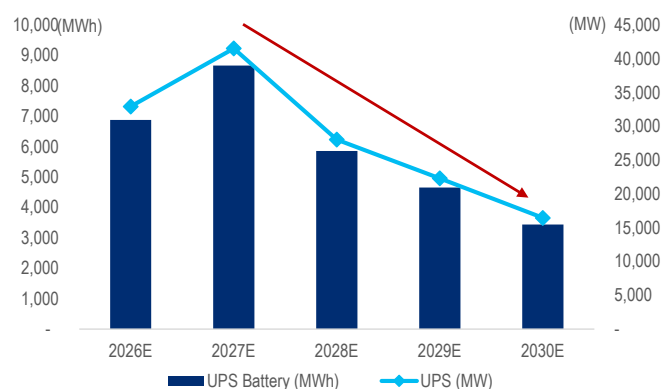
BESS operates at the facility level, providing minute-scale backup power (replacing the centralised UPS), peak shaving of large cluster-level load ramps, and optionally grid frequency response services. The combined architecture represents a fundamental shift from centralised, AC-domain backup toward distributed, DC-native energy management — creating durable demand for LFP battery cells, battery management systems, and DC-coupled power conversion equipment across the value chain.

Figure 12. BESS and BBU Become Important Parts in 800VDC Architecture



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Source: Citi Research Estimates

Figure 13. ...While UPS to Be Replaced in SST Architecture



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Source: Citi Research Estimates

We believe UPS capacity demand — which remains elevated in 2026–27E as the sidecar retrofit phase retains legacy grey-space UPS infrastructure — should enter a sustained downtrend from 2028 as facility-level SST builds eliminate the UPS from the primary power path entirely. This decline will not be cyclical but structural within the 800VDC architectural paradigm.

Key Beneficiaries across Infrastructure Stack within a Data Center

We map the physical and functional anatomy of an AI data center in Figure 14, from site infrastructure to rooftop mechanical systems, and identify the key equipment categories that should stand to benefit from the structural transition to 800VDC. We organise the investment implications across six major dimensions.

Data Center Operators: DC operators are the primary demand aggregators of this transition. Their campuses house every layer of the architecture illustrated from MV switchgear intake through to server halls, and their ability to deliver high-density, power-ready data centers to hyperscaler tenants is the central commercial proposition. When AI rack densities escalate and 800VDC becomes the delivery standard in the future, operators with flexible grey-space design and power infrastructure upgrade capability should command premium lease rates and stronger tenant retention.

Power Equipment: The power infrastructure layer — comprising MV switchgear/RMU, transformers, backup generators, UPS systems, and PUD/RPP — is the most directly disrupted segment. Under 800VDC, centralised AC UPS and LV transformers are progressively retired, replaced by upstream MV rectifiers and ultimately SST. This represents both displacement risk for incumbent AC equipment vendors and a significant new market for DC-native power conversion equipment manufacturers.

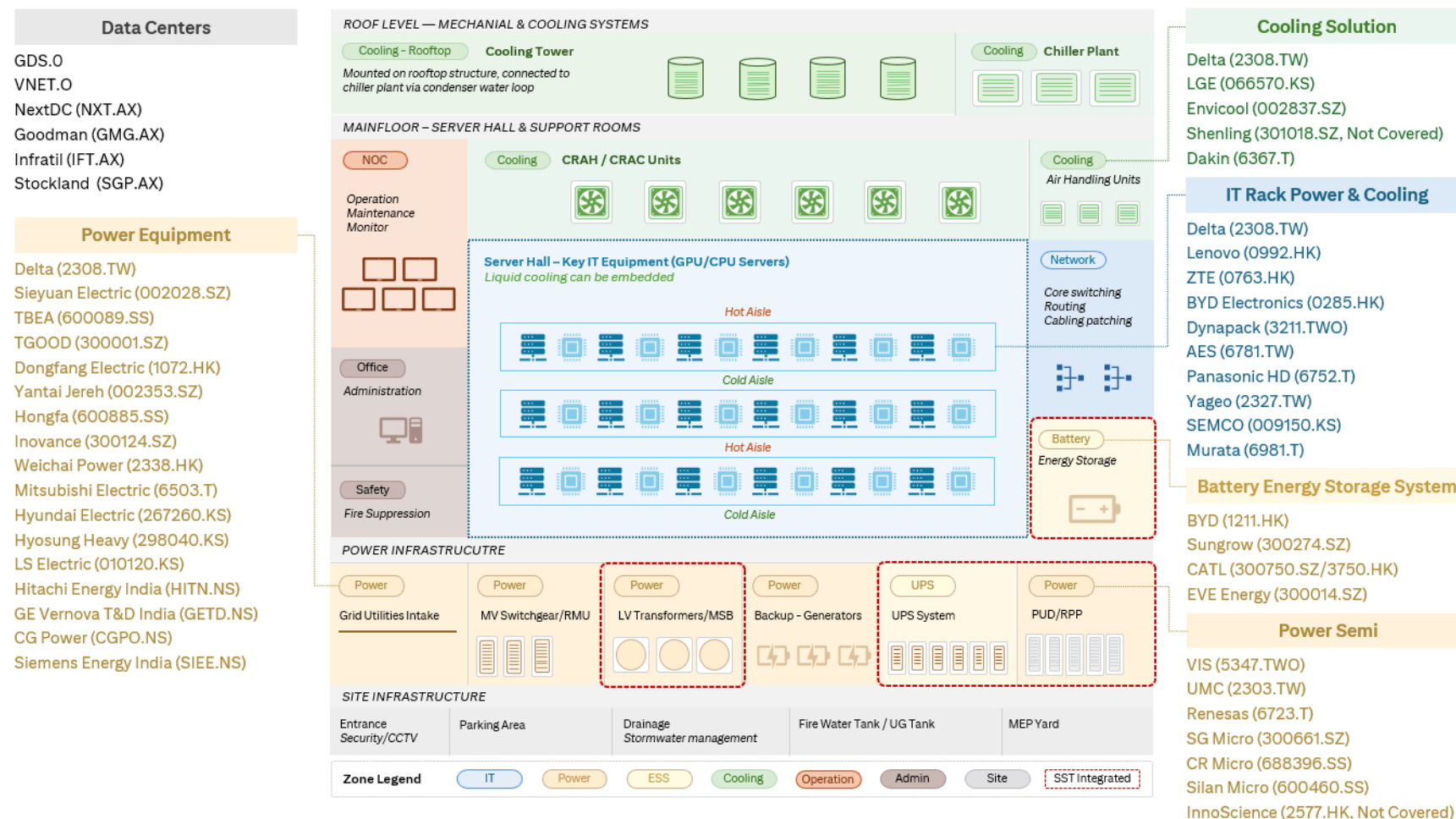
Battery Energy Storage Systems: The facility-level BESS becomes structurally more important under 800VDC, replacing the centralised UPS as the primary power resilience mechanism. China's dominant position in LFP battery manufacturing translates into a direct supply chain advantage, in our view. It is a meaningful revenue opportunity for battery system integrators serving the data center vertical.

IT Rack Power and Cooling: The GPU upgrade cycle is the primary demand driver for IT rack infrastructure including rack-level BBU. As successive GPU generations (Blackwell, Rubin, Kyber) deliver step-change improvements in compute density and performance, the associated rack system's power and cooling related components require corresponding upgrades. Vendors supplying high-density rack enclosures, power whips, and structured cooling solutions are incremental beneficiaries of this generational refresh cycle.

Cooling Solutions: Higher rack power densities enabled by 800VDC directly necessitate a parallel upgrade in thermal management. As per-rack power crosses the 100kW threshold, traditional air-cooling architectures become thermally insufficient. Liquid cooling solutions become standard deployment requirements rather than optional upgrades. Cooling solution providers with established data center verticals should benefit from this co-investment cycle alongside the power architecture transition.

Power Semiconductors: The shift to high-voltage DC conversion is a direct demand catalyst for wide-bandgap power semiconductors. SSTs, MV rectifiers, and high-efficiency DC/DC converters all rely on silicon carbide (SiC) and gallium nitride (GaN) devices to achieve the switching frequencies and efficiency levels required at 800VDC. This positions SiC/GaN device manufacturers as a key enabling layer of the entire 800VDC ecosystem.

Figure 14. Data Center Architecture Layout and Key Asia Beneficiaries for 800VDC Transition



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Source: Citi Research

Deep Dive into Key Asia Beneficiaries in 800VDC Transition

We address the key Asia beneficiaries in data center operation, power grid and plant equipment, battery energy storage system, IT rack power and cooling, data center cooling solutions, semis and passive components in this section.

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In the China data center market, we like **GDS** and **VNET**. We see the transition as a positive demand catalyst in the near term, with manageable capex implications in the medium term, underpinned by the flexibility of their existing grey-space infrastructure and the competitive advantage of a mature domestic equipment supply chain.

AI demand remains the primary catalyst. We view that hyperscaler cluster rental pricing has stabilised and is showing early signs of reacceleration into 2H26, underpinned by sustained growth in LLM training and inference workloads, supporting accelerating EBITDA delivery for both GDS and VNET as committed pipelines convert to revenue in the coming years. The demand story remains intact as we outlined in our [June's China Data Centers report](#).

800VDC commercial rollout in China is still in early stage. Domestic AI chip platforms (Huawei Ascend, domestic alternatives) have not yet been standardised on 800VDC power delivery, and rack power densities in China's AI compute ecosystem remain below the thresholds that are driving urgent 800VDC adoption. Thus, we believe there is minimal near-term capex pressure for GDS and VNET.

Capex impact likely manageable with a China supply chain advantage. Based on SemiAnalysis' phase analysis, the net capex differential of 800VDC versus traditional AC infrastructure is modest. Chinese IDC operators benefit from additional cost advantages: domestic LFP battery manufacturers provide best-in-class BESS at globally competitive prices, while domestic power equipment vendors offer structural pricing advantages. We believe GDS and VNET should be able to execute any future 800VDC transition at a meaningful discount to global capex benchmarks.

Grey-space design provides built-in upgrade flexibility. Well-designed grey-space infrastructure is compatible with row-level sidecar retrofit without structural rebuild. When domestic tenant demand for 800VDC materialises, GDS and VNET could respond with targeted, low-disruption upgrades on a campus-by-campus basis, avoiding stranded asset risk.

In Australia, NextDC (NXT AX, Buy) and CDC—owned by Infratil (IFT AX, Buy)—are among the leading DC operators. Both companies develop and operate data centers across Australia and New Zealand, serving hyperscalers, cloud providers, and colocation customers. Recent contract wins have lifted contracted capacity to roughly **700MW for NextDC** and **1GW for CDC**, reflecting strong demand for AI and cloud infrastructure. Both operators continue to expand their development pipelines, and future facilities could incorporate **800V DC power architecture** to improve sellable capacity and enhance energy efficiency.

Among Australian real estate names, Goodman Group (GMG AX, Buy) and Stockland (SGP AX, Buy) offer growing exposure to the DC theme through large development pipelines. Both are investing in DC projects, either directly or through partnerships, with Stockland developing assets alongside **EdgeConneX** in a 50/50

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joint venture. For both companies, the key value driver is likely to be development profit rather than recurring operating income, given their broader and diversified property portfolios. Goodman is advancing a global **6.4GW power bank** with more than **500MW of near-term projects**, and data centers could become a major earnings contributor, rising from around **10% of revenue in FY26 to 50–60% by FY29–30E**. Stockland currently has no meaningful earnings exposure but could see data centers contribute **2–3% of earnings between FY28 and FY30**, with further upside possible from future developments. As these projects are built out, both developers could incorporate **800V DC power architecture** to improve sellable capacity.

Power Equipment

Power Grid Equipment

The Asia power grid equipment sector is supported by strong global demand, particularly for high-voltage transformers and switchgear serving the robust capital expenditure cycle in AC data-center development.

We highlight major companies by country:

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Taiwan: Delta Electronics (2308 TW, Buy) has successfully transformed from a traditional power supply manufacturer into a comprehensive AI infrastructure provider, offering solutions that span the entire power delivery chain from grid to chip. Delta's portfolio covers AC/DC conversion, HVDC power systems, rack-level power infrastructure, battery backup solutions, and liquid cooling technologies. This end-to-end capability positions the company uniquely within the AI data center ecosystem, as few competitors can provide integrated solutions across both power and thermal management. As data center operators increasingly prioritize efficiency, reliability, and deployment speed, Delta's ability to deliver a complete power architecture solution should become an increasingly important competitive advantage.

AI-related infrastructure, including power systems, power management equipment, and thermal solutions—is estimated to contribute 25–30% of Delta's total revenue in 2026, making it the company's largest growth engine. We expect this contribution to increase further as hyperscalers and cloud service providers accelerate the buildout of next-generation AI clusters. While near-term deployments are likely to focus on $\pm 400\text{V}$ HVDC architectures, we anticipate initial shipment volumes to begin ramping in 2H26. The more meaningful inflection point should occur from 2027 onward, when 800V DC systems begin broader commercialization alongside the scaling of Nvidia's Rubin platform and other next-generation AI infrastructure deployments.

As rack power density continues to increase, the industry's transition toward higher-voltage architecture appears less a technology upgrade and more a structural necessity. Given Delta's leadership across the power infrastructure stack and its expanding exposure to AI power systems, we view the company as one of the direct and compelling beneficiaries of the multi-year transition to 800V DC data center architectures.

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Korea: All major transformer and switchgear players, such as **Hyundai Electric (267260 KS, Buy)**, **Hyosung Heavy (298040 KS, Buy)** and **LS Electric (010120 KS, Buy)**, are getting more orders related to 800VDC applications. Regarding SST development, Hyosung Heavy developed 22.7kV SST technology back in 2022; for 800V DC architecture and data center applications, the company is in pilot and demonstration stage with a US hyperscaler; this phase is expected to be completed within 2026E. For Hyundai Electric, it is in an R&D phase for SST and has a three-year target. It has secured LV transformer orders for datacenter projects this year, which will be delivered from 2029. For LS Electric, it has developed prototype that

has been installed on an in-house test bed, though it is difficult to specify the schedule of commercial sales.

China: Sieyuan Electric (002028 CH, Buy) is a leading China high-voltage transformer and switchgear manufacturer; it has received orders for the former from data centers in the US. We believe China companies have competitive advantages in terms of lead time and production costs versus foreign peers. **Sungrow (300274 CH, Buy)** has launched new product for power distribution systems at AIDC; new orders are expected to be received in 2H26E and shipment would likely be made in 2027E. **TGOOD (300001 CH, Buy)** launched an AIDC integrated power supply product in June 2026, which can transform 110kV to 200–800V power supply for domestic customers, but it could also be an emerging LV transformer player in the future as it has a competitive cost structure with rising market share in non-US markets. **Inovance (300124 CH, Buy)** and **Hongfa (600885 CH, Buy)** have exposure to SST and power relay products.

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Although a relatively late entrant to the space, **Inovance** has begun developing SST products targeting 800V DC AI data center applications. The company recently announced a collaboration with HCCCAP, a company incubated by Tsinghua University's carbon nanomaterials laboratory, to develop integrated power supply solutions for both AIDC and ESS. The partnership leverages HCCCAP's high-power pulse supercapacitor technology, which is designed to deliver power density of up to 100kW/kg. The solution is designed to provide millisecond-level response capabilities, enabling transient power support, peak-load buffering, and fail-safe backup function features that could become increasingly important as AIDC moves toward higher-voltage and higher-power-density architectures. That said, we believe commercialization remains at an early stage and therefore expect only limited revenue contribution from Inovance's 800V DC AIDC-related products in 2026–27. While we think the company is taking the right strategic steps to build exposure to next-generation power infrastructure, meaningful financial benefits are unlikely to materialize until broader adoption of 800V DC architectures gains traction over the longer term.

Hongfa is already a key supplier of relay products for AIDC infrastructure, with current exposure primarily through automatic transfer switches (ATS). As the industry transitions toward 800V DC power architectures, however, the company's addressable market should broaden significantly.

We expect Hongfa to supply a more comprehensive range of relay products across multiple power infrastructure components, including power distribution units (PDUs), battery backup units (BBUs), solid state transformers (SSTs), and other high-voltage power management systems. Such expansion would increase both content per deployment and the company's participation across the AI data center power chain.

According to management, revenue from 800V DC AI data center relay products could reach Rmb1.0–1.5bn in 2026. This estimate is based on several key assumptions: (1) relay content of approximately Rmb80,000–100,000 per MW of data center capacity, (2) global data center capacity reaching around 40GW, (3) 800V DC penetration rising to roughly 50%, and (4) Hongfa maintaining an approximately 50% market share in the relevant relay segment.

Commercialization could begin in earnest in 2H26, when Hongfa starts supplying 800V HVDC relay products to North American end customers through power infrastructure partners, such as Delta Electronics and other power management solution providers.

Reflecting the growing adoption of AI infrastructure and next-generation power architectures, we expect AIDC-related revenue to increase from approximately 4-5% of total sales in 2025 to 8-10% in 2026E, before rising further to the mid-teen percentage range in 2027E. As 800V DC deployments scale, Hongfa's expanding product portfolio and strong position in the relay market should enable it to capture an increasing share of the power infrastructure value chain.

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Japan: Mitsubishi Electric (6503 T, Buy) plans to expand its high-voltage transformer production capacity in Japan, North America and Southeast Asia. The expansion will be primarily through upgrading existing facilities plus partial expansion, not large-scale, completely new built ones. Mitsubishi Electric's cumulative high-voltage transformer production capacity, from the base in 2024, would be +15% in 2025, +30% in 2026E, +45% in 2027E and +50% in 2028E by spending US\$193m globally.

India: We believe India's DC market is approaching a critical inflection point, with accelerating demand, supportive government policies, and large-scale infrastructure expansion all converging. Global technology leaders, including Microsoft, Amazon, and Meta, have collectively announced commitments of approximately **US\$67.5 billion** toward AI and data center infrastructure in India, highlighting the country's growing importance within the global digital economy. Based on publicly disclosed company roadmaps, announced, planned, and early-stage capacity additions now exceed **8GW**, with a majority expected to come online by 2032.

We believe India's leading transformer and T&D OEMs are well positioned to benefit from this buildout. **Hitachi Energy India (HITN NS, Buy)**, **GE Vernova T&D India (GETD NS, Buy)**, **CG Power and Industrial Solutions (CGPO NS, Buy)**, and **Siemens Energy India (SIEE NS, Neutral)** have historically focused on domestic transmission projects and intra-group exports. However, all four companies are now actively expanding into the DC segment, representing a meaningful broadening of their addressable markets.

Hitachi Energy India estimates that its current addressable opportunity in AI-driven DC infrastructure accounts for roughly **10-15% of total DC capex**. Through its "Grid-to-Rack" strategy, management aims to expand this exposure by an additional **15-20%**, potentially increasing its addressable share to **25-35% of total AI DC infrastructure spend**. To support this strategy, the company has announced a **Rs10 billion capex program** focused on localizing technologies tailored to India's AI DC requirements.

GE Vernova T&D India has already established an early track record in the DC market through supplies made to its parent company for US DC projects. This provides the company with valuable technical credentials and reference projects as it seeks to expand its presence in the domestic DC market.

CG Power and Industrial Solutions entered the DC export market in January 2025, securing a **Rs9 billion order** linked to a US DC project. We view this as an important validation of the company's DC product capabilities and a positive milestone as it positions itself for a larger role in India's rapidly expanding DC ecosystem.

Siemens Energy India is also expanding its DC product portfolio, with an increasing focus on solutions that support the growing power requirements of next-generation AI and hyperscale facilities.

With all four major India T&D OEMs actively broadening their DC offerings, we see DC-related TAM expansion as a key structural growth driver for the sector over the

coming years. The combination of accelerating domestic capacity additions, rising AI infrastructure investment, and increasing localization of DC supply chains should create a meaningful long-term opportunity for India's power equipment leaders.

Power Plant Equipment

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On power plant equipment, we are constructive, as the order mix shifts increasingly towards non-coal generation. Representatives in this space include **Yantai Jereh Oilfield Services (002353 SZ, Buy)**, which benefits from the rapid expansion of AIDC in the US, capitalizing on two critical market shortages — power supply and gas turbines. Unlike many peers, Jereh has proactively secured a diversified and reliable gas turbine supply base, sourcing from industry heavyweights including GE Vernova, Baker Hughes, Siemens, and Kawasaki. This strategic procurement advantage has enabled Jereh to leverage its robust gas turbine power generator inventory to win incremental AIDC orders, positioning it ahead of competitors constrained by the prevailing gas turbine supply deficit.

Beyond equipment supply, Jereh is deeply embedded in the DC supply chain, offering integrated "Gas Turbine + Energy Storage" bundles and operating as a full EPC turnkey provider and equipment manufacturer for power integrated services. In early May, Jereh further strengthened its supply chain resilience by establishing a joint venture with FTAI, converting retired CFM56 jet engines into gas turbine power generators — securing yet another stable and cost-effective turbine source while gaining a unique entry point into the high-growth data center power market.

Dongfang Electric (1072 HK, Buy) should also stand to benefit meaningfully from rising demand for gas turbines from its North American datacenter customer.

In the power generation/backup segment, **Weichai Power (2338 HK, Buy)** provides critical onsite/backup power solutions for data centers. Its gas engines support high-power requirements in hybrid setups alongside 800VDC systems, ESS, and grid connections—essential for reliability in AI infrastructure buildout. Its products also provide an alternative to backup systems.

Power generation for AI data centers should be a key growth driver for Weichai going forward. We expect power generation for AI data centers to contribute 5%/8%/12% of its total revenue and 21%/38%/43% of total earnings in 202E/27E/28E. Further reading: [Weichai Power \(2338.HK\) - Powering up more AIDCs; Lift up 26-28E NP by 4/13/56% and TP to HK\\$50](#)

Mgmt. expects 2–3 MW of gas engines to be launched in mid-2026 with mass delivery in 3Q/4Q26, and 5 MW of gas engines to see prototype & validation by end-2026 ([link](#)). As for profitability of gas engines (2–3 MW), mgmt. indicated their GPM is better than diesel engines GPM (35–40%), given higher ASP of US\$600k/MW versus diesel engine's Rmb1mn/MW. Gas engines production can share core engine line with diesel engines. The production line does not need large-scale reconstruction, with low conversion cost and strong flexible production capacity.

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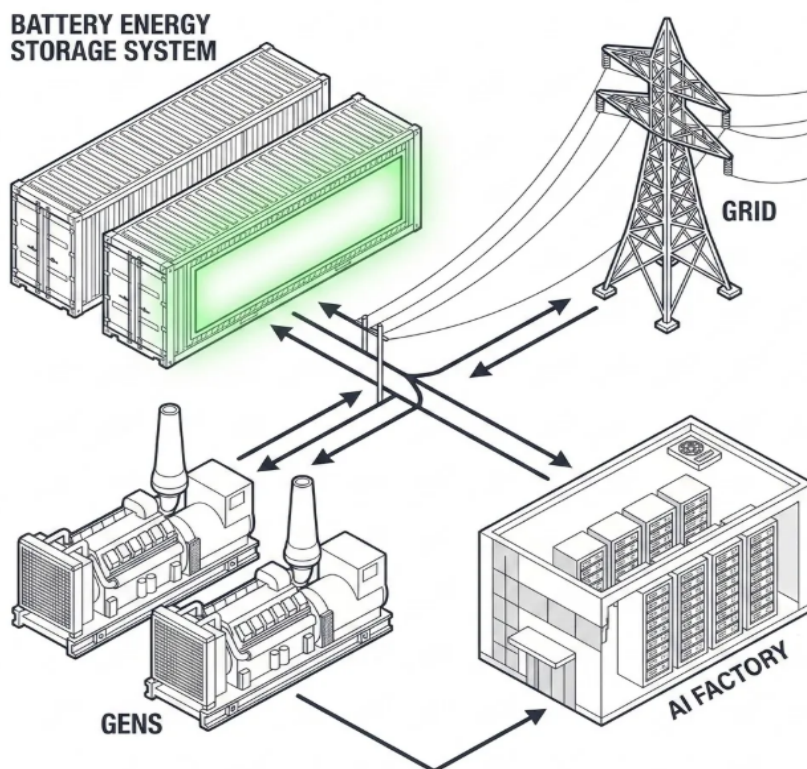
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Battery Energy Storage System

Battery energy storage systems (BESS) are quickly becoming essential infrastructure for AI factories. Unlike traditional data centers, AI factories are built to manufacture intelligence at scale. They run power-dense training and inference workloads, increasingly support agentic and reasoning models, and must deliver predictable performance even as compute demand shifts rapidly. In NVIDIA DSX, the platform for AI factories, BESS is part of the broader AI factory power architecture rather than a standalone add-on. Properly designed BESS can help AI factories connect faster, operate more reliably, reduce stress on the grid and onsite generation, and manage the fast-changing load profiles created by large-scale AI workloads.

BESS is an integrated system that combines battery cells with power conversion systems (PCS) inverters, advanced telemetry, and dynamic control schemes. The batteries store energy, but the inverters and controls make the system grid-interactive, shaping how power is absorbed, injected, and regulated in real time. BESS is a smart, controllable power asset, not a passive energy reservoir.

Figure 15. BESS design for AI factories



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Source: Nvidia, Citi Research

1GW of data centers for AI computing would need approximately 3.3GWh of BESS for backup power supply for grid-friendly data centers, according to Sungrow. The company projected that this emerging application could add 10–20GWh to BESS demand in the US in 2026, and demand could double yoy in 2027E. We see Sungrow as a key beneficiary as the company is one of the leading ESS suppliers in the US with 10–20% market share. At present, energy storage is

being integrated into data center projects primarily because it grants priority in the project queue—projects equipped with storage can proceed ahead of others. Large-scale deployment of energy storage systems in data centers is likely to start from 2027 after the industry forms unified industry standards.

Among energy storage and inverter manufacturers we want to highlight: **Sungrow (300274.CH, Buy)**, **BYD (1211.HK, Buy)**, **CATL (300750.SZ/3750.HK, Buy)**, and **EVE Energy (300014.SZ, Buy)**.

Sungrow is supported by broadening product demand — including the launch of its new SST product line in July 2026 — as the US market transitions to 800VDC architecture. Sungrow planned to launch its 10kV SST product on 9 July 2026. This is a mature, market-ready version designed for immediate deployment based on extensive discussions held with potential clients across Europe, the US, and the domestic market. SST is a programmable, digital, and intelligent power supply component, which can drive the power electronics industry's evolution from the analog and digital eras. The commercialization of SST is based on the essential technical foundation including the rapid advancements in high-frequency transformer materials and the SiC power device technology as well as comprehensive upgrades in fiber optics, DSP chips, and insulation technologies.

Sungrow possesses technical expertise in the fields of power electronics and new energy, thereby enabling the industrialization of SSTs. Sungrow's SST application is not limited to data center usage, it will also be applied to EV charging piles, energy storage, solar and wind power, and DC microgrids, etc.

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BYD is a leading player in energy storage systems and battery solutions. It supplies high-safety LFP blade batteries, supporting DC-coupled systems, peak shaving, backup power, grid stabilization, and integration with high-voltage DC architectures in AI/data center environments.

Energy storage is a high-growth segment within BYD's broader new energy business. According to mgmt., ESS revenue accounted for around 10% of total BYD revenue (>60GWh shipment) in 2025, which we anticipate to rise to 12%/15% of total revenue (150/200GWh) in 2027E/28E. Further reading: [BYD \(1211.HK\) - May/Jun inventory estimate; 2Q-3Q earnings re-rating conditions almost mature, in our view](#)

BYD continues to roll out large-scale ESS solutions (e.g., high-capacity containerized systems) tailored for reliable power in data centers and digital infrastructure. It has a strong track record in global deployments, with a focus on safety, high power density, and integration for modern DC-heavy environments.

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CATL, as one of the leading players in ESS and battery solutions, is also pivoting from a pure ESS provider to a vertically integrated energy infrastructure solution provider for AI data centers, establishing an ecosystem to capitalize on the significant power demand from AI data centers in the long term.

Energy storage system business laid the foundation of the ecosystem – CATL has been dedicated in the ESS space for years, from pioneering LFP battery for ESS, to pursuing larger battery size from 314Ah to 587Ah, and now to launching its first validated sodium-ion energy system, showcasing its ambition against the backdrop of AI-driven power demand surges.

To midstream power conversion – CATL invested approximately US\$600mn for a 49% stake in Zhongheng Technology in April 2026, and therefore a ~17.4% stake in subsidiary Zhongheng Electric, to leverage Zhongheng's HVDC and solid-state

transformer strengths. Zhongheng Electric is a leading Chinese power solutions provider deeply integrated into the AI data center ecosystem driven by Nvidia, and its HVDC power technologies are used to cool and power next-generation GPU racks.

Going further downstream – Associated CATL funds reportedly agreed to buy up to a 38.1% stake in data center operator VNET Group, with a value of up to US\$942mn ([Bloomberg](#), May 13, 2026). We view that the strategic acquisition indicates CATL's ambition to secure a foothold in the AIDC and data center infrastructure space.

EVE Energy is also one of the pioneers of energy storage systems, and ESS has been a major contributor to its business, contributing ~40% of topline revenue in 2025. EVE Energy keeps gearing up the R&D process and ventures into the rising AI space with competitive ESS battery products; it unveiled its comprehensive battery backup units (BBU) in June 2026 to cater for the differentiated demand emerging from AI data centers.

IT Rack Power and Cooling

Battery Backup Unit

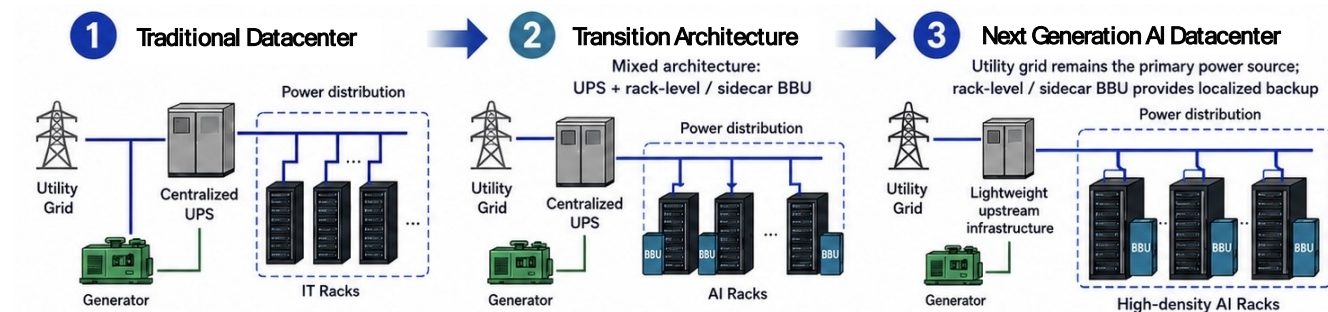
BBU becomes increasingly essential under HVDC architecture: We believe the transition toward 800V HVDC architecture could further strengthen the importance of BBU in next-generation AI data centers. Under traditional AC power architectures, centralized UPS remains the primary backup solution, requiring multiple AC/DC and DC/AC conversion stages before power reaches IT equipment. By contrast, HVDC distributes high-voltage DC directly to the rack, making battery-based backup a more ideal extension of the rack-level DC power path.

As AI rack power increases from today's 100–150kW toward several hundred kilowatts in future platforms, power continuity becomes increasingly critical. A brief interruption can disrupt the high-value GPUs operating within a rack. We therefore believe backup power needs to move closer to the load to minimize conversion losses, improve ride-through response and simplify rack-level power delivery. Under HVDC architecture, BBU can be integrated directly into the DC distribution system, reducing reliance on multiple power conversion stages while improving overall system efficiency.

Importantly, we believe HVDC not only increases BBU adoption but could also drive up its content value. Higher rack power requires larger battery capacity, higher discharge power, more complex battery management system (BMS), thermal management, protection, and high-voltage safety design. We expect future BBU products are likely to evolve from relatively simple battery modules into integrated high-power subsystems with significantly greater content and system value.

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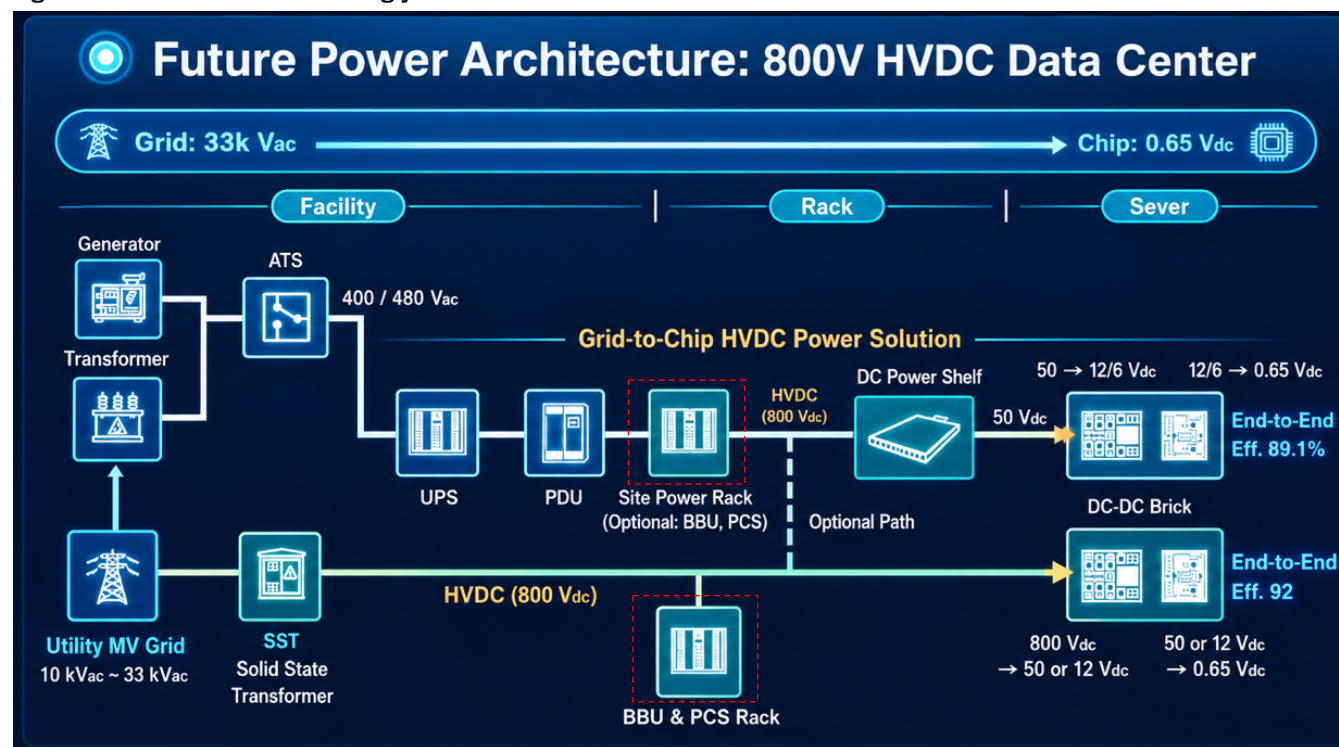
Figure 16. Increasing Adoption of BBU in AIDC



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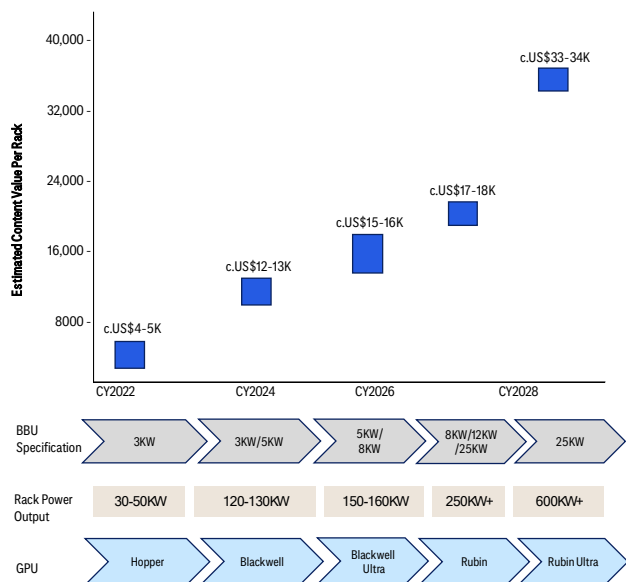
Source: Citi Research

Figure 17. BBU Becomes Increasingly Essential under HVDC Power Architecture



Source: Delta

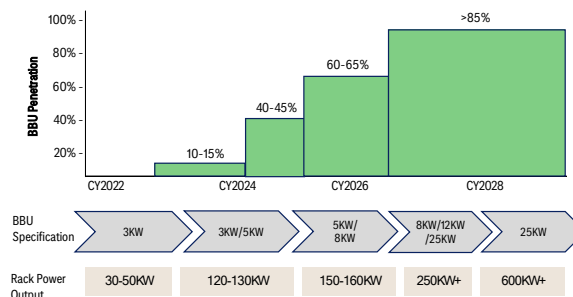
Figure 18. Estimated Content Value Per Rack by Generation



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Source: Citi Research, Citi Research Estimates

Figure 19. BBU Penetration Based on Our Estimates



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Source: Citi Research, Citi Research Estimates

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Japan: Panasonic (6752 T, Buy) is a high-market-share supplier of battery backup units (BBUs) for servers and is already conducting R&D on high-output BBUs designed for dedicated power supply racks — a key component of the emerging 800VDC architecture. The company plans to be ready for mass production by FY3/27. In parallel, Panasonic expects to begin mass production of capacitor backup units (CBUs), which utilise supercapacitors to absorb temporary peak loads, also within FY3/27.

As a medium-term target, Panasonic has guided for ¥950 billion in sales across its combined BBU/CBU business by FY3/29, of which approximately ¥200 billion is expected to be contributed by next-generation products, including high-output BBUs and CBUs.

We estimate that data center-related businesses — including BBU/CBU — could account for more than 20% of Panasonic's adjusted operating profit by FY3/28, a rising earnings mix that we believe could help drive a re-rating of the stock.

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Taiwan: Dynapack (3211 TT, Buy) has long-standing experience in lithium-ion battery pack design, battery management systems, safety certifications, and customized manufacturing. We believe BBU provides a meaningful growth opportunity beyond its mature consumer electronic battery-pack business and the company is well positioned to benefit from rising BBU demand driven by AIDC infrastructure buildout.

We estimate BBU sales likely contributed 35% of Dynapack's group sales in 2025 and could accelerate to 55%/67% of group sales in 2026E/2027E. We forecast Dynapack's sales to rise at a 34% CAGR in 2025-28E, boosted by the ongoing ramp-up of higher-power BBU products, rising rack power density leading to higher content value and penetration of BBUs, and the company's share gains.

A majority of products sold to PSU operators are 3KW and 5.5KW BBU modules in 2026 and the company expects to start selling small volume of higher-KW products (8.5KW, 12.4KW) in 4Q26. The products for HVDC (25KW BBU) could begin mass production in 2027, further fueling topline growth. Further reading: [Taiwan Electronic Components & Equipment – Riding the AIDC BBU Growth Cycle: Initiate Dynapack at Buy.](#)

AES (6781 TT, Buy) is also a direct beneficiary of AIDC BBU adoption, in our view. BBU now contributes >70% of sales and should remain the key earnings driver as AI rack power continues to rise. We expect AES to benefit from higher BBU content per rack and deep CSP engagement as the industry transitions from conventional 48V systems to HVDC architecture.

We expect high-voltage (+/-400V and 800V) BBU solutions to enter volume production in early 2027, supported by next-generation AI racks with higher power density. The transition raises technical barriers, which should favor established suppliers with proven system integration capability. We estimate AES's sales to rise at a 24% CAGR in 2025-28E.

Rack Cooling

Lenovo (0992 HK, Buy) and **ZTE (0763 HK, Neutral)** should benefit from IT rack cooling requirement upgrade, driven by higher power density requirement amid the 800VDC transition. We see increasing participation of China's electronic supply chain to join the Nvidia 800VDC ecosystem, especially the MGX platform, to provide components for power systems and cooling solutions.

We see 800VDC adoption and the associated rack density and thermal management requirements it drives as potential multi-year tailwinds for **Lenovo's** ISG order intake. Lenovo has invested meaningfully in Neptune liquid cooling technology, its proprietary direct liquid cooling (DLC) architecture, which is already deployed at scale at several hyperscalers' data centers globally. As per-rack power densities cross the 100kW threshold and approach 600kW+ with Blackwell and future GPU generations, Neptune-equipped rack solutions should become a prerequisite, rather than a premium option — expanding Lenovo ISG's addressable market and improving its solution ASPs. The shift toward differentiated, high-density AI rack systems with integrated liquid cooling represents a meaningful mix-shift opportunity toward higher-margin, solution-oriented revenue for Lenovo, in our view.

ZTE's data center business supplies high-density IT rack enclosures, integrated power distribution modules, and liquid cooling units to domestic cloud and telco data center customers. The 800VDC transition reinforces demand across all three product lines simultaneously: higher-density GPU deployments drive IT cabinet upgrades; the shift from AC UPS to DC-native power delivery creates replacement demand for ZTE's modular power systems; and rising rack thermal loads necessitate liquid cooling co-deployment. ZTE's established relationships with China's major telco operators provide a natural distribution channel for its integrated data center infrastructure products.

Lead Wealth, under BYD Electronics (285 HK, Sell), is one of the MGX electronica ecosystem companies for 800VDC data centers. It provides power system components such as power shelf, cold plate, UQD, CDU, manifold, PDP, busbar, etc. The company can leverage its precision mechanical capability and CNC capacity to support the ramp. But currently, contribution is still small.

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MLCC

We believe passive components are poised to benefit from a structural increase in content intensity. Unlike power semiconductors, capacitors, resistors and inductors are required throughout every stage of the power delivery chain—from front-end EMI filtering and surge protection, through intermediate DC/DC conversion, to final power delivery network (PDN) stabilization near the processor. The shift to 800VDC introduces higher power density, more conversion stages, and tighter power integrity requirements, resulting in increased passive component usage per rack. As next-generation AI clusters scale in power consumption and computing density, we expect passive content per compute unit to rise meaningfully, creating a broad-based and recurring demand driver across the passive component ecosystem.

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Murata (6981 JT, Buy) is the global leader in MLCCs and we believe it is increasingly well positioned to benefit from the AI data center power buildout. Beyond its dominant MLCC franchise, the company is expanding into next-generation power solutions, including **vertical power delivery (VPD)** technology, which brings power conversion closer to the processor to improve efficiency and reduce power loss. Murata also offers **intermediate bus converters (IBCs)** and **front-end AC-DC power solutions**, giving it exposure across both board-level components and power delivery systems.

We estimate Murata holds roughly **50% share of the AI data center MLCC supply chain**, while its power business is entering early stages of commercialization. As adoption of next-generation power architecture accelerates, we believe Murata should be well positioned to expand its presence in power infrastructure alongside its established MLCC leadership. We expect AI data center applications to account for approximately **25% of MLCC revenue in FY26** and **38% in FY27**, up from around **12% in FY25**. Meanwhile, the power business remains small at about **¥28bn in revenue** (less than 2% of sales) but is approaching a meaningful growth phase as VPD and other advanced power solutions move into volume production.

Overall, we continue to view Murata as the best-positioned player in the MLCC industry, supported by its technology leadership, market share, and growing exposure to next-generation AI power infrastructure. A key catalyst to watch is the pace of adoption of its power modules, particularly VPD-based solutions.

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In Korea, **Samsung Electro-Mechanics (009150 KS, Buy)**, leveraging its position as the world's second-largest MLCC manufacturer, is benefiting from rapidly growing demand for both AI-grade MLCCs and embedded MLCC solutions as hyperscalers and cloud service providers scale next-generation AI infrastructure.

As AI servers become increasingly power-intensive, the number and complexity of MLCCs required per system continue to rise. The transition to higher-voltage, higher-power-density architectures is driving greater demand for high-capacitance, high-voltage, and high-reliability MLCCs, creating a structural content increase per server and per rack. We believe this trend will serve as a powerful long-term demand driver for Semco's high-specification passive component portfolio, positioning the company as one of the most direct beneficiaries of the AI-driven upgrade cycle in data center power infrastructure.

We forecast Semco's MLCC ASP to increase by approximately 30% YoY in 2026E and a further 71% YoY in 2027E. This growth should be supported by sustained AI server demand, a richer product mix, and increasing MLCC content requirements across next-generation 800VDC power architectures. The transition to higher-voltage and higher-power-density systems is driving demand for more advanced

MLCCs with enhanced capacitance, voltage tolerance, and reliability characteristics.

Supply conditions for AI-grade MLCCs are also expected to tighten through 2027E, reflecting strong AI infrastructure investment and a limited number of suppliers capable of producing the specialized components required for AI servers. We believe the industry is entering a period of increasing supply-demand imbalance as hyperscalers and OEMs accelerate deployment of next-generation AI clusters.

Semco appears particularly well positioned to benefit from this trend, in our view. Customer interest remains strong for the company's newly launched 47 μ F AI server MLCCs, while demand is also rising for high-voltage MLCCs rated above 1kV, which are increasingly required to improve power conversion efficiency in 800VDC systems. In addition, the transition toward 800Gbps and 1.6Tbps optical interconnects is driving demand for ultra-high-capacitance MLCCs exceeding 100 μ F, further expanding content opportunities for Semco across the AI infrastructure value chain.

Overall, we believe Semco's technology leadership in high-value MLCC products, combined with the industry's migration toward 800VDC power architectures, should position the company as one of the most direct beneficiaries of the next wave of AI data center infrastructure investment.

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In Taiwan, **Yageo (2327 TT, Buy)** offers one of the most comprehensive exposure to the 800VDC power architecture. Through its leading positions in MLCCs, tantalum capacitors, resistors, and magnetic components, the company participates across virtually every power conversion and power distribution stages, providing a breadth of exposure that few single-product peers can replicate. At the high-voltage front end, Yageo is benefiting from strengthening demand for high-reliability NPO MLCCs, where industry supply remains tight. Based on our industry checks, lead times have extended and key customer orders are likely to resume in 2H26. At the processor-level PDN layer, Yageo's KEMET tantalum capacitors remain a preferred solution among major hyperscaler customers due to their superior reliability and power efficiency characteristics. This business also represents one of the company's highest-margin product segments. Meanwhile, at intermediate power conversion stages, Yageo's high-capacitance MLCC portfolio should be well positioned to capture incremental demand as several Japanese and Korean competitors increasingly allocate production capacity toward premium AI-grade and automotive products, creating opportunities for market share gains in mainstream and mid-range applications.

While Yageo does not separately disclose revenue attributable to 800VDC infrastructure, the broad deployment of passive components throughout the power delivery chain creates a portfolio-wide demand uplift. AI-related revenue contribution has already increased from approximately 4% of total sales in early 2025 to around 15% in 1Q26, reflecting accelerating adoption of AI server infrastructure.

Looking further ahead, we estimate that next-generation AI rack architectures in 2027-28 could drive more than a 50% increase in MLCC unit consumption per rack, accompanied by substantially higher value content due to stricter performance specifications. In parallel, tantalum capacitors are expected to become an increasingly important growth driver, with tantalum-related revenue contribution potentially exceeding 25% of total sales over time.

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The tantalum market backdrop remains supportive, in our view. Ongoing supply disruptions have pushed tantalum ore prices higher, leading Yageo to implement three rounds of price increases since 2025. With supply-demand conditions remaining tight, we believe additional pricing actions could follow in 2H26, providing a further tailwind to margins and earnings growth.

Cooling Solutions

In Korea, **LG Electronics (066570 KS, Buy)** should benefit from the 800VDC transition through its growing exposure to AIDC cooling infrastructure.

LGE's initial focus is on supplying air-cooling solutions, including its proprietary high-efficiency chillers, to AIDC. These products could serve as the company's primary entry point in the rapidly expanding AI infrastructure market. Over time, LGE plans to commercialize liquid cooling systems, which are increasingly viewed as a necessity for next-generation AI clusters as rack power densities continue to rise beyond the practical limits of conventional air cooling.

This roadmap positions LGE to progressively expand its exposure across the data center thermal management value chain, from traditional HVAC systems to advanced liquid cooling solutions that support future 800VDC AI architectures.

The opportunity builds upon LGE's established strengths in the HVAC market. According to management, HVAC revenue has increased approximately fourfold since 2001, while commercial and industrial HVAC revenue has grown roughly 30-fold over the same period, reflecting the company's successful expansion into higher-value enterprise applications.

Supported by a global manufacturing footprint of 12 production sites, LGE can provide end-to-end HVAC solutions, encompassing product development, manufacturing, installation, and after-sales service. This global and integrated service capability should enhance its competitiveness as hyperscalers and data center operators increasingly seek reliable partners for large-scale AI infrastructure deployments.

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In China, we highlight **Envicool (002837 SZ, Sell)** as one of China's leading liquid cooling solution providers. The company has established a comprehensive liquid cooling portfolio covering full-chain liquid cooling systems, immersion cooling solutions, coolant distribution units (CDUs), and related thermal management infrastructure designed for next-generation AI data centers and Nvidia-based AI server deployments.

The transition toward 800VDC architecture could accelerate liquid cooling adoption. As rack power density rises beyond the capability of conventional air-cooling systems, liquid cooling becomes increasingly necessary to managing higher thermal loads, while maintaining power efficiency and system reliability. This structural shift should create meaningful demand opportunities for Envicool's liquid cooling offerings over the coming years.

However, despite the favorable industry backdrop, we maintain our Sell rating. We estimate the liquid cooling business will contribute only a mid-single-digit to high-single-digit percentage of total revenue in 2026, up from approximately 3% in 2025. While growth is impressive, the segment remains relatively small compared with the company's overall revenue base.

In addition, competitive intensity is rising rapidly. Domestic players such as Shenling (301018 SZ, non-rated) are aggressively expanding their presence in

liquid cooling solutions for Nvidia AI servers, which could pressure both market share and pricing over time. As a result, we view liquid cooling primarily as a volume growth story for Envicool rather than a significant margin expansion opportunity.

Notably, profitability within the liquid cooling segment has yet to demonstrate a clear advantage over the company's traditional air-cooling business. Gross margins of liquid cooling products were broadly comparable to those of air-cooling solutions last year, suggesting limited operating leverage despite strong demand growth. Execution also remains a concern. Envicool missed earnings expectations for three consecutive quarters, from 3Q25 through 1Q26, raising questions over the pace at which the company can translate industry tailwinds into consistent financial performance.

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In Japan, **Daikin (6367 T, Buy)** is a key AI infrastructure name to watch. The company expects revenue from the North American DC cooling market to more than triple to over **¥300 billion (US\$1.9 billion)** by 2030, supported by direct sales to hyperscalers such as AWS and Google, which typically carry higher margins. Even if the industry's transition to **800VDC power architecture** progresses more slowly than expected, we believe Daikin remains well positioned to achieve its targets. Recent acquisitions, including **Chillydyne**, have strengthened its liquid-cooling capabilities, with direct-to-chip cooling solutions expected to deliver a **25%+ sales CAGR** over the next five years.

Looking ahead, an important industry trend is the shift toward more efficient power architectures in data centers. Cooling accounts for more than **60% of non-IT power consumption**, yet no major vendor currently offers a fully **DC-native cooling system**. For now, this supports strong demand for power-management components such as inverters. However, the emergence of DC-native cooling solutions could eventually disrupt parts of the existing inverter supply chain.

Power Semiconductors

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As addressed in our US tech team's note ([US Semiconductors - 800V DC Transition Drives Stronger Analog Recovery](#)), the shift to high-voltage DC conversion is a direct demand catalyst for wide-bandgap power semiconductors. SSTs, MV rectifiers, and high-efficiency DC/DC converters all rely on silicon carbide (SiC) and gallium nitride (GaN) devices to achieve the switching frequencies and efficiency levels required at 800VDC. This positions SiC/GaN device manufacturers as a key enabling layer of the entire 800VDC ecosystem.

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We see 800VDC adoption driving demand for: 1) high-voltage power discrete and modules (SiC MOSFET, GaN, SJ MOSFET, and 2) PMIC and analog IC, especially for multi-phase controllers, DrMOS, high-current DC/DC, and more. CR Micro estimates (citing TrendForce) the AI server power market to grow from US\$7.4bn in 2025 to US\$32.5bn in 2027 (CAGR of 110%), leading to significant growth for power semiconductors. While international CSPs typically prefer non-Chinese semiconductor vendors in their supply chain, we believe China's domestics market is sufficiently large, accounting for roughly one-third of global server shipment, with meaningful localization potential. We see **SG Micro (300661.SZ, Buy)**, **CR Micro (688396.SS, Neutral)** and **Silan Micro (600460.SS, Neutral)** as key beneficiaries.

We believe **SG Micro** stands to benefit from the industry shift towards 800VDC given its growing product portfolio for 1) server power management: high-power DC/DC, eFuse, DrMOS, 2) networking: TEC drivers, AFE, 3) signal chain: ADC/DAC, OpAmps. Content value could reach several US\$ per optical module and US\$30-

50 per server. Computing & networking revenue contribution rose from 12% in 2024 to 15% in 2025, and likely 20% in 1Q26. We expect the strong demand to become an increasingly important growth driver for the company. We also believe **CR Micro** and **Silan Micro** could benefit from accelerating SiC/GaN and high-voltage MOSFET demand, leading to improving capacity utilization and product mix.

As noted in our recent Suzhou tech tour takeaways, InnoScience is a [domestic GaN supplier](#). Management's estimate of ~100,000 GaN chips per MW of data center power — at a data center ASP of US\$1.0–2.0 per unit — directly ties InnoScience's revenue trajectory to the 800VDC buildout cycle. The company's 8-inch Si-based GaN production line has a self-assessed 2–3 year lead over 6-inch peers and it targets 70k/month capacity by end-2027/early-2028. Although management estimates China's domestic 800VDC adoption to lag international peers, it aims to further amplify 800VDC revenue contribution through the planned mix shift from modules toward higher-margin wafers and chips.

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We see Taiwanese foundries as key beneficiaries of the transition toward 800V HVDC data center architectures. Higher operating voltages are driving increased demand for advanced BCD technologies, alongside broader adoption of GaN and SiC process platforms to improve power efficiency and address increasingly stringent thermal management requirements.

Within this evolving power infrastructure, **Vanguard (5347 TT, Buy)** and **UMC (2303 TT, Buy)** occupy critical positions in the semiconductor manufacturing stack that converts high-voltage grid power into stable computing voltages for AI servers and data centers. While VIS is primarily exposed to the front-end of power distribution and conversion, UMC is more leveraged to downstream power management functions closer to the processor.

We believe **VIS** is particularly well positioned at the high-voltage end of the power chain. Leveraging its mature-node manufacturing platform, the company supplies key power semiconductor components such as high-voltage isolation ICs, hot-swap controllers, and power management devices used in power supplies and rack-level power conversion systems. More importantly, VIS is one of the few foundries offering a comprehensive power process portfolio across a broad voltage range. Through its licensing agreement with TSMC for 80V and 650V GaN technologies, combined with its proprietary GaN-on-QST platform, VIS supports both silicon-based and GaN manufacturing, providing solutions spanning sub-200V, 650V, and up to 1,200V applications. This positions the company as a one-stop manufacturing partner for next-generation high-efficiency power devices across the data center power architecture.

The transition to higher-voltage power distribution should drive growing content opportunities for PMICs and other power devices. We estimate revenue contribution from data center-related PMICs could increase from a single-digit percentage of VIS's sales in 2025 to a high-teen percentage in 2026, with further expansion over the following years as AI infrastructure deployment accelerates. In parallel, VIS's Singapore expansion, which plans to lift capacity to approximately 44,000 wafers per month by 2029, should provide additional room to capture rising demand for high-voltage power semiconductors and advanced power management applications.

Moving further downstream in the power architecture, voltage is progressively stepped down through multi-phase voltage regulator modules (VRMs) before reaching the processor. At this stage, the critical challenge shifts from high-voltage

power conversion to precise power regulation and distribution. These functions rely on highly integrated PMICs that combine analog, logic, and power devices on a single chip, requiring sophisticated BCD and embedded high-voltage process technologies.

This is where **UMC** is strategically positioned. The company is leveraging its leading BCD and embedded high-voltage platforms to enable next-generation PMICs that improve power efficiency within an increasingly dense and power-intensive AI computing environment. As demand for PMICs rises alongside the adoption of 800V HVDC architectures, UMC should see increasing opportunities in specialty-node manufacturing. The company is also expanding capacity in Singapore with a focus on 22/28nm and specialty technologies, strengthening its ability to support the growing market for power management and high-power semiconductor applications.

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In Japan, **Renesas Electronics (6723 T, Neutral)** currently supplies products for data centers that integrate power management ICs and power semiconductors. In recent years, this area has been a major driver of revenue growth for the data center segment. Renesas announced a product compatible with 800VDC architecture. Specifically, it announced a 400V-compatible product that uses GaN, gallium nitride, as a power semiconductor. This enables support for 800V by using multiple units. The GaN technology is derived from Transphorm, which was acquired in 2024. The company held its Capital Market Day in Jun-25, 2026 and commented that the AI infrastructure business, including the digital power business, could account for just under 20% of total company revenue in 2026 and c40% by 2030, thereby positioning AI infrastructure as the company's growth driver for the foreseeable future.

Glossary

Abbreviation	Full Term	Brief Introduction
AC	Alternating Current	The standard form of electrical power delivered by the utility grid, oscillating in direction at 50/60Hz
ATS	Automatic Transfer Switch	A device that automatically switches power supply between the primary grid source and backup generator
BCD	Bipolar-CMOS-DMOS	A semiconductor process technology combining multiple transistor types, used for power management ICs
BESS	Battery Energy Storage System	A facility-level energy storage system providing backup power, peak shaving, and grid frequency response
BBU	Battery Backup Unit	A rack-level energy storage device providing millisecond-to-second buffering of GPU load transients
BMS	Battery Management System	Electronic system monitoring and controlling the charging, discharging, and safety of battery cells
CBU	Capacitor Backup Unit	A rack-level device using supercapacitors to absorb sub-100ms power transients, complementing BBU
CDU	Coolant Distribution Unit	Infrastructure component distributing liquid coolant to server racks in liquid cooling deployments
DC	Direct Current	Unidirectional electrical current; the native power form for compute chips and battery storage
DLC	Direct Liquid Cooling	A thermal management technique circulating liquid coolant directly to heat-generating components
DrMOS	Driver + MOSFET	An integrated power stage combining gate driver and power MOSFET in a single package for DC/DC conversion
DSP	Digital Signal Processor	A specialized processor for high-speed signal processing, used in power conversion control systems
GaN	Gallium Nitride	A wide-bandgap power semiconductor material enabling high-efficiency, high-frequency power conversion
HVAC	Heating, Ventilation and Air Conditioning	Traditional thermal management systems for data centers, increasingly supplemented by liquid cooling
HVDC	High-Voltage Direct Current	DC power distribution at elevated voltage levels (in this report, primarily 800VDC)
IBC	Intermediate Bus Converter	A DC/DC power conversion component stepping down intermediate bus voltage for board-level power delivery
kVA	Kilovolt-Ampere	Unit of apparent (total) electrical power; common rating unit for transformers and generators
LFP	Lithium Iron Phosphate	A lithium-ion battery chemistry favored for energy storage applications due to safety and cycle life
LV	Low Voltage	Electrical systems operating below 1,000VAC or 1,500VDC, covering standard data center distribution
MLCC	Multi-Layer Ceramic Capacitor	A passive electronic component used extensively throughout power conversion and signal filtering circuits
MOCVD	Metal-Organic Chemical Vapor Deposition	The manufacturing process used to grow GaN epitaxial layers on silicon wafers
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor	A fundamental power switching device used in DC/DC converters and power management circuits
MSB	Main Switchboard	The primary LV electrical distribution board receiving transformer output and routing power downstream
MVA	Megavolt-Ampere	Unit of apparent power used to rate transformers, SSTs, and generators at facility scale
MV	Medium Voltage	Electrical systems operating between 1kV and 35kV, covering utility grid interconnection to data centers
NOC	Network Operations Centre	The facility control and monitoring hub for data center operations
NVLink	NVIDIA high-speed interconnect	NVIDIA's proprietary high-bandwidth GPU interconnect enabling large-scale synchronous GPU domains
PDN	Power Delivery Network	The on-board network of passive components delivering stable voltage to processor cores
PDU	Power Distribution Unit	Equipment distributing conditioned AC power to rack rows with metering and circuit protection
PCS	Power Conversion System	The inverter/converter component of a BESS managing bidirectional energy flow between battery and grid

PFC	Power Factor Correction	A power electronics function ensuring efficient real power draw and minimizing reactive current from the grid
PMIC	Power Management Integrated Circuit	A chip integrating multiple power regulation and conversion functions for computing platforms
PSU	Power Supply Unit	The in-rack device performing AC-to-DC conversion to power server boards and GPUs
PUE	Power Usage Effectiveness	A data center efficiency metric equal to total facility power divided by IT load power; target is 1.0
RPP	Remote Power Panel	A sub-distribution panel extending PDU power routing to individual rack rows
RMU	Ring Main Unit	A compact MV switching device used for grid interconnection and protection at data center intake
SiC	Silicon Carbide	A wide-bandgap power semiconductor material enabling high-efficiency conversion at elevated voltages
SOC	State of Charge	The remaining energy capacity of a battery expressed as a percentage of its total rated capacity
SST	Solid-State Transformer	An advanced power conversion device replacing the LV transformer by converting MV AC directly to 800VDC
SSCB	Solid-State Circuit Breaker	A semiconductor-based protection device replacing mechanical breakers in DC distribution systems
TDP	Thermal Design Point	The maximum sustained power a GPU or processor is designed to dissipate under reference conditions
UPS	Uninterruptible Power Supply	A centralised power conditioning and backup system; progressively displaced by distributed BBU/BESS in 800VDC
VAC	Volts Alternating Current	Unit expressing AC voltage level (e.g., 415VAC, 480VAC)
VDC	Volts Direct Current	Unit expressing DC voltage level (e.g., 800VDC, 54VDC)
VPD	Vertical Power Delivery	A next-generation power architecture bringing voltage conversion physically closer to the processor die
VRM	Voltage Regulator Module	A DC/DC converter providing precise regulated voltage to processor cores from intermediate bus voltage

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Appendix A-1

ANALYST CERTIFICATION

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The Firm has made a market in the publicly traded equity securities of Contemporary Amperex Technology Co Ltd on at least one occasion since 1 Jan 2025.

The Firm has made a market in the publicly traded equity securities of Lenovo Group Ltd on at least one occasion since 1 Jan 2025.

The Firm has made a market in the publicly traded equity securities of ZTE Corp on at least one occasion since 1 Jan 2025.

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